

Day: M Conceptual Short Question No: 3

How does the terminal speed of a parachutist before opening parachute compare to the terminal speed afterward? Why is there a difference?

The terminal velocity of a parachutist before opening the parachute is different from its velocity after opening the parachute.

Reasons and Explanation

- The dragging force depend upon the exposed surface area of falling object i.e greater the exposed surface area of the body, greater will be the dragging force and low will be its terminal velocity and vice versa.
- When the parachutist is falling down without opening parachute, the exposed surface of parachutist to air resistance is small. So he experiences less drag force than his weight and falls with high terminal velocity.
- When the parachutist is falling down with open parachute, large surface area is exposed to air resistance.
- As a result the parachutist experience large drag force and his terminal velocity is low.